THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS VIII — MATHEMATICS · MID-TERM PRACTICE PAPER

Session 2026-27 | CBSE / RBSE Pattern

Class: 8	Subject: Mathematics	Set 02	Time: 2 Hours 30 Minutes	Max. Marks: 60

General Instructions:

1. This question paper contains 16 questions. All questions are compulsory.
2. Question paper is divided into FIVE sections – Section 'A', 'B', 'C', 'D' and 'E'.
3. In Section 'A' question number 1 has 12 multiple choice sub-parts carrying 1 mark each.
4. In Section 'B' question numbers 2 to 7 are very short answer type questions carrying 2 marks each.
5. In Section 'C' question numbers 8 to 10 are short answer (SA) type questions carrying 3 marks each.
6. In Section 'D' question numbers 11 to 13 are long answer (LA) type questions carrying 5 marks each.
7. In Section 'E' question numbers 14 to 16 are case-based questions whose sub-parts are of 1, 1 and 2 marks respectively.
8. There is no overall choice. However, internal choice has been provided in 1 question of Section B, 1 question of Section C, 2 questions of Section D, and in the 2-mark sub-part of every case-based question in Section E. Attempt only one of the alternatives in all such questions.
9. Draw neat figures wherever required. Use of calculators is NOT allowed. Write the serial number of the question before attempting it.

SECTION A — Multiple Choice Questions (12 × 1 = 12) — 1 questions · 12 marks

1. Question 1 consists of multiple-choice questions (i–xii) of 1 mark each. Choose the correct option.

(i) The number of non-square natural numbers between 9^2 and 10^2 is: **1**

(a) 17 (b) 19 (c) 20 (d) 18

(ii) Which of the following is a perfect cube? **1**

(a) 1800 (b) 2025 (c) 1728 (d) 3000

(iii) The value of 2^{-4} is: **1**

(a) 1/16 (b) −16 (c) −8 (d) 1/8

(iv) $(3^5)^2 \div 3^7$ equals: **1**

(a) 3 (b) 27 (c) 9 (d) 81

(v) The standard form of 6,50,000 is: **1**

(a) 65×10^4 (b) 6.5×10^4 (c) 6.5×10^5 (d) 0.65×10^5

(vi) 449 written in Roman numerals is: **1**

(a) CDXLIX (b) CCCXLIX (c) CDLIX (d) CDXLVIIII

(vii) The base-3 number 212 equals which decimal number? **1**

(a) 21 (b) 22 (c) 23 (d) 25

(viii) A quadrilateral with exactly one pair of parallel sides is a: **1**

(a) Parallelogram (b) Trapezium (c) Rhombus (d) Rectangle

(ix) In an isosceles trapezium ABCD with AB ∥ DC, if ∠A = 70°, then ∠D is: **1**

(a) 70° (b) 110° (c) 90° (d) 100°

(x) Kaprekar's constant for 4-digit numbers is: **1**

(a) 6174 (b) 495 (c) 1089 (d) 9999

(xi) $(a + b)^2 - (a - b)^2$ equals: **1**

(a) 2ab (b) $2a^2 + 2b^2$ (c) 0 (d) 4ab

(xii) If 4 pencils cost ₹ 30, the cost of 10 pencils is: **1**

(a) ₹ 70 (b) ₹ 80 (c) ₹ 75 (d) ₹ 120

SECTION B — Very Short Answer (6 × 2 = 12) — 6 questions · 12 marks

Questions 2 to 7 carry 2 marks each.

2. Find the smallest number by which 72 must be multiplied to get a perfect cube. **2**

3. Evaluate: $(2^{-1} + 3^{-1})^{-1}$ **2**

4. Convert $(110101)_2$ into a decimal number. **2**

5. Without actual division, check whether 234567 is divisible by 9. **2**

OR

Find the digit * so that 3*76 is divisible by 11.

6. The diagonals of a rectangle are (3x + 5) cm and (5x − 7) cm. Find x and the length of each diagonal. **2**

7. Divide ₹ 540 between Meena and Neha in the ratio 4 : 5. **2**

SECTION C — Short Answer (3 × 3 = 9) — 3 questions · 9 marks

Questions 8 to 10 carry 3 marks each.

8. Convert 69 into (i) base 5 and (ii) base 3. **3**

9. In a rhombus ABCD, ∠A = 70°. Find ∠B, ∠C and ∠D. State the property used. **3**

OR

In trapezium PQRS, PQ ∥ SR, ∠P = 75° and ∠R = 110°. Find ∠S and ∠Q.

10. Expand and simplify: **3**

(i) $(2a + 3)(a^2 - 4a + 5)$ (ii) $(3x - 2y)(x + 5y)$

SECTION D — Long Answer (3 × 5 = 15) — 3 questions · 15 marks

Questions 11 to 13 carry 5 marks each.

11. (i) Find the smallest number by which 1176 must be multiplied so that the product is a perfect square. Find the square root of the product. **5**

(ii) The area of a square field is 2704 m². Find the length of its side.

OR

(i) Find the smallest number by which 2000 must be divided so that the quotient is a perfect cube. Find the cube root of the quotient.

(ii) The volume of a cube is 4096 cm³. Find the length of its edge.

12. If $17^2 = 289$, find: (i) 1.7^2 (ii) 0.17^2 (iii) 170^2 (iv) 1700^2 (v) 0.017^2 **5**

OR

Simplify and write in exponential form: (i) $5^{-2} \times 5^7$ (ii) $2^{-3} \div 2^{-5}$ (iii) $(9^{-3})^2$ (iv) $(3/4)^{-2}$ (v) $p^6q^2 \times (pq)^{-2}$

13. (i) Apply Kaprekar's routine to the 3-digit number 352 and show every step until it reaches 495. **5**

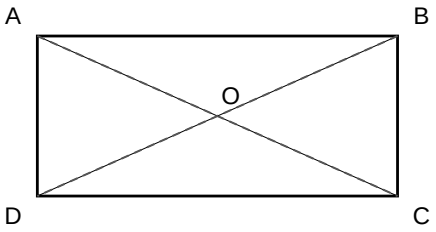
(ii) What happens if the routine is applied to 111? Why?

(iii) Apply the routine to 2025 (4-digit) and state the number of steps taken to reach 6174.

SECTION E — Case-Based Questions (3 × 4 = 12) — 3 questions · 12 marks

Questions 14 to 16 carry 4 marks each (1 + 1 + 2).

14. A rectangular herb garden ABCD has a diagonal path AC = 26 cm on the school's model plan. The diagonals AC and BD meet at O, and the side AB = 10 cm.



(i) Find the length of OB. 1

(ii) Find the length of the side BC. 1

(iii) Find the perimeter and the area of the rectangular garden. 2

OR

(iii) A string is laid along both diagonals at ₹ 45 per cm on the model. Find the total cost.

15. In a laboratory a culture starts with 5×10^4 cells, and the number of cells becomes 10 times larger every day.

(i) How many cells will there be after 3 days? Give your answer in standard form. 1

(ii) Write the number of cells after 3 days in usual form. 1

(iii) Find the ratio of the number of cells after 2 days to that after 5 days. 2

OR

(iii) After how many days will the culture first exceed 5×10^9 cells?

16. A school canteen prepares a sherbet by mixing lemon juice, sugar and water in the ratio 1 : 2 : 7 to make 5 litres of sherbet in each batch.

(i) How many litres of sugar are used in one batch? 1

(ii) Write the ratio of lemon juice to water in simplest form. 1

(iii) How much water is needed along with 2.4 litres of lemon juice (same ratio)? 2

OR

(iii) Lemon juice costs ₹ 60 per litre and sugar ₹ 40 per litre. Find the cost of the ingredients for one 5-litre batch (water is free).